

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Amanda Padua
Program/Batch	B083C-Networkwalks
Date	25 September 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM4 (theHarvester)
Client/Target	1. NetworkWalks – Microsoft is used for an educational footprinting exercise using theHarvester. No active exploitation was performed. 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Kali Linux tools and Setup Phase 3-5: Practical Footprinting and Gathering of Information

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

For this project, I had the opportunity to get some hands-on experience with **footprinting and reconnaissance using Kali Linux**. The main goal was to understand how information about a company can be collected during the early stages of a cybersecurity assessment.

I worked with several tools, including **theHarvester**, **nslookup**, **WHOIS**, **curl**, **WhatWeb**, and **wafw00f**. I also used **Baidu** as a search source with theHarvester. Each tool gave me a different type of information, which helped me understand how reconnaissance works in practice.

What I found useful about this exercise was being able to actually run the commands myself and look at the results rather than only learning about these tools in theory.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux	The main environment I used for the practical exercise.
theHarvester	Gather publicly available email addresses, sub-domains, host and other domain- related information
Baidu	Used as the search source with theHarvester
nslookup	Perform DNS lookups and view DNS related information
whois	Look up publicly available domain registration information
curl -I	View the HTTP response headers returned by the target website.
WhatWeb	Identify technologies and components associated with the target website.
wafw00f	Check for indications of a Web Application Firewall (WAF).

4. Activities Performed

4.1 Setting Up the Kali Linux Environment

I started by opening **Kali Linux** and preparing the Terminal for the practical activities. I made sure I could access the required tools before starting the reconnaissance tasks.

4.2 Using theHarvester

I used **theHarvester** to perform passive information gathering on the assigned target, microsoft.com.

The purpose was to see what information could be found from publicly available sources, such as **email addresses, sub-domains, and hosts**.

This gave me a better understanding of how information gathering can be carried out without directly accessing the target's internal systems.

4.3 Footprinting with Baidu

For the first theHarvester task, I used **Baidu** as the search source and set the result limit to **1,000**.

After running the command, I went through the results and looked at the information that was returned, including email addresses, sub-domains, and other domain-related information.

I also took screenshots of the command and results so I could use them as evidence in my report.

4.4 Footprinting Using All Sources

For the second theHarvester task, I performed another search against microsoft.com, but this time I selected all available sources.

I set the result limit to 50 and reviewed the information returned by the tool.

Doing this helped me see that using different sources can produce different results and gave me more experience working with theHarvester.

4.5 DNS Information Using nslookup

I used nslookup to perform a DNS lookup for the target domain, microsoft.com.

This allowed me to view DNS-related information from the command line and helped me understand how domain names can be connected to DNS records.

Using nslookup gave me a more practical understanding of DNS reconnaissance rather than only learning about it theoretically.

4.6 DNS Information Using nslookup

I used **WHOIS** to look up publicly available information related to the **microsoft.com** domain.

This helped me understand the type of domain registration information that can be viewed during the reconnaissance stage.

4.7 Using curl -I

I used curl -I to send a request to the target website and view the HTTP response headers returned by the web server.

This gave me some practical experience with using command-line tools to examine how a website responds to a request and what information can be included in its HTTP headers.

I also captured the output as evidence for the project.

4.8 Identifying Website Technologies with WhatWeb

I used WhatWeb to gather information about technologies and components associated with the target website.

This helped me understand how reconnaissance can be used to learn more about the technologies behind a publicly accessible website.

4.8 Checking the Website with wafw00f

I used **wafw00f** to check the target website for indications of a **Web Application Firewall (WAF)**.

This was useful for understanding how reconnaissance can also involve identifying security technologies that may be protecting a web application.

5. Risk Analysis / Impact

After going through the results from the different tools, I noticed that a fair amount of information about the target could be found through publicly available sources. These findings do not mean that the target is vulnerable, but they show how information gathered during reconnaissance could potentially be useful for further investigation.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Public email and sub-domain information available	theHarvester returned publicly available email addresses, sub-domains, and host-related information	This information could potentially be used for further reconnaissance, phishing, or social-engineering attempts	● Medium
2	DNS information available	nslookup returned DNS-related information for the target domain	This could help someone understand more about the target's public-facing infrastructure	● Low
3	Domain registration information available	WHOIS returned publicly available information related to the target domain	This provides additional information that could be useful during the reconnaissance stage	● Low
4	HTTP response headers exposed	curl -I returned HTTP response headers from the target website	Some header information may help with identifying technologies or carrying out further enumeration	● Low
5	Website technologies identifiable	WhatWeb identified technologies and components associated with the target website	Knowing the technologies being used could help someone identify software that may need further security review	● Medium
6	WAF information identifiable	wafw00f provided information about whether a Web Application Firewall could be identified	This may give someone additional information about the security technologies protecting the website	● Low

Risk level key: ● Critical ● Medium ● Low

From this exercise, I found that even without trying to access the target's internal systems, quite a lot of information could be collected through **publicly available sources**.

The findings do not automatically mean that there is a security vulnerability. However, when different pieces of information are combined, they can give someone a better understanding of an organization's online infrastructure.

Because of this, regularly checking what information is publicly exposed can help organizations reduce unnecessary information disclosure and make reconnaissance more difficult for potential attackers.

6. Recommendations

Based on what I found while working through the different reconnaissance activities, I would recommend the following improvements:

1. **Review publicly available information**
Organizations should regularly check what information about their domains, email addresses, sub-domains, and websites can be found publicly. This can help reduce unnecessary information exposure.
2. **Keep web technologies updated**
Websites and the technologies behind them should be kept up to date. Regular updates can help reduce the risk of known security issues being left unaddressed.
3. **Review HTTP response headers**
HTTP response headers should be checked regularly to make sure they are not giving away more technical information than necessary about the website or server.
4. **Review DNS records regularly**
DNS records should be checked from time to time to make sure only the required domains, hosts, and services are publicly available.
5. **Review domain registration information**
Organizations should be aware of what information about their domains can be found through public WHOIS records and review whether any unnecessary information can be limited.
6. **Properly configure and monitor the WAF**
If a Web Application Firewall is being used, it should be kept updated and monitored regularly. Its configuration should also be reviewed to make sure it is providing the intended protection.
7. **Monitor publicly exposed technologies**
Organizations should keep track of the technologies being used on their public-facing websites. Any unexpected or outdated technology should be investigated and reviewed.
8. **Provide cybersecurity awareness training**
Employees should receive regular awareness training about phishing and social engineering. This is especially important when company email addresses and other contact information are publicly available.
9. **Perform security testing with authorization**
Reconnaissance and security testing should only be carried out on systems where proper authorization has been given. Regular authorized testing can help organizations understand what information is exposed and where improvements can be made.

7. Conclusion

Overall, this project was a really useful hands-on experience for me because I got to actually work with different Kali Linux reconnaissance tools instead of just learning about them in theory.

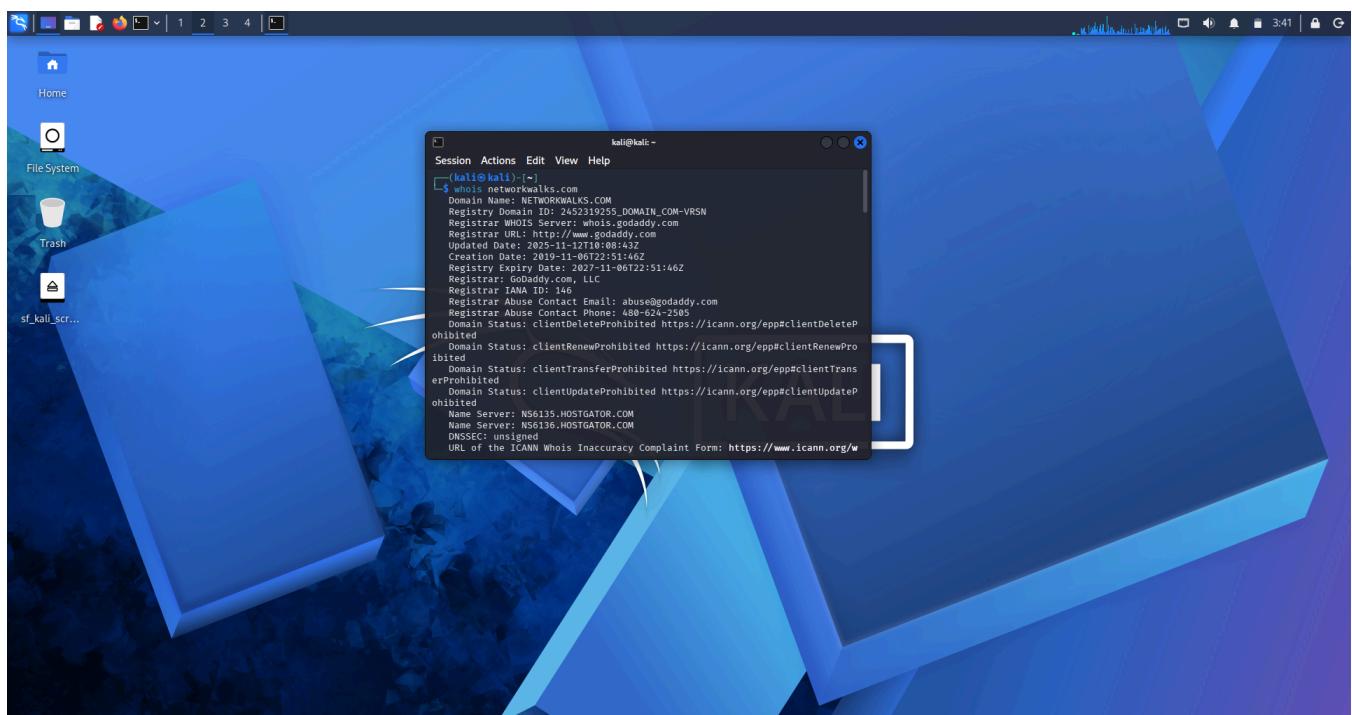
I learned that each tool has a different purpose. **theHarvester** helped me look for publicly available email addresses and sub-domains, while **nslookup** and **WHOIS** helped me understand DNS and domain information. I also used **curl -I** to look at HTTP response headers, **WhatWeb** to learn more about the technologies used by the website, and **wafw00f** to check for information about a possible WAF.

One of the things that stood out to me the most was how much information can be found from **publicly available sources** without needing to access a company's internal systems. This helped me understand why reconnaissance is an important part of a cybersecurity assessment.

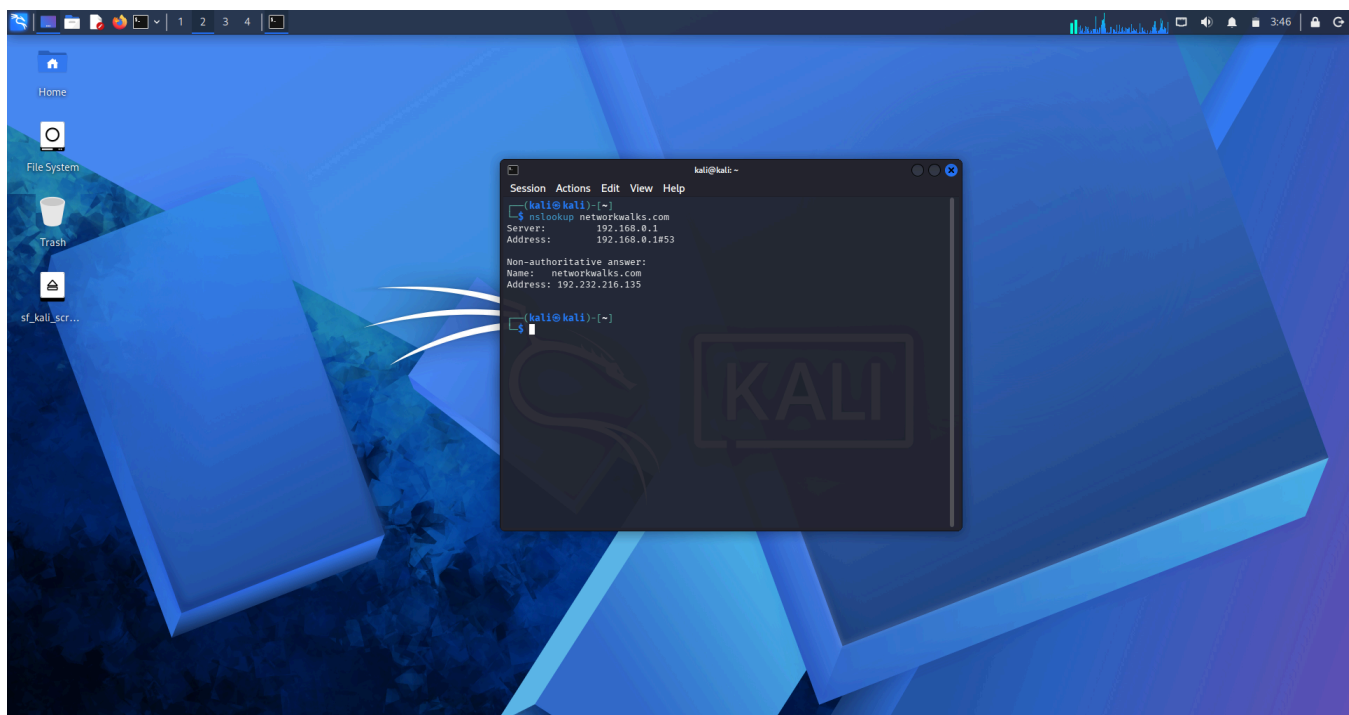
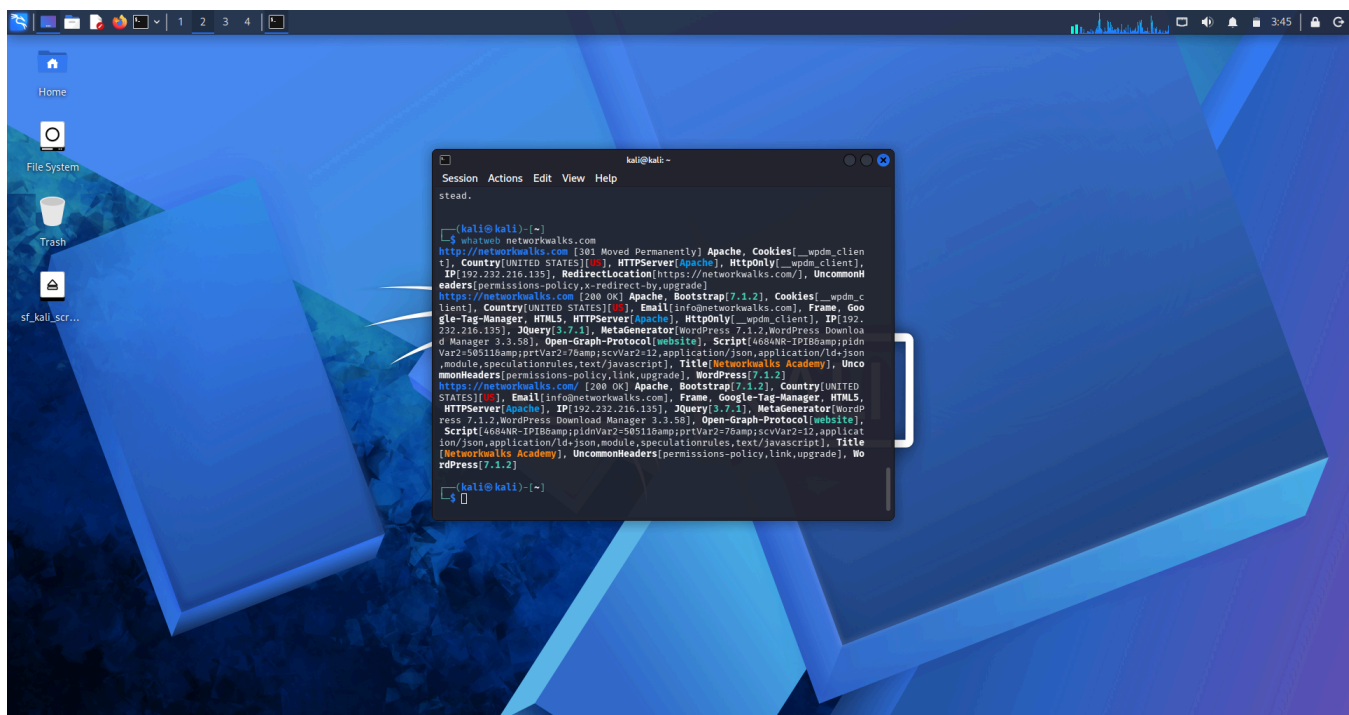
I also became more comfortable using the Kali Linux Terminal and running different commands. Overall, this project helped me put what I have been learning into practice and gave me a better understanding of how the reconnaissance stage works in a real cybersecurity assessment.

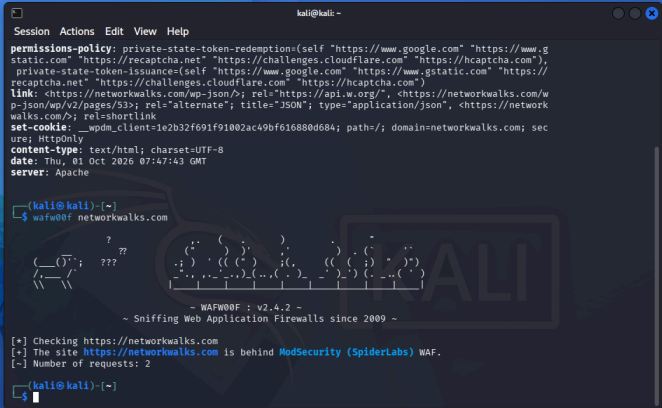
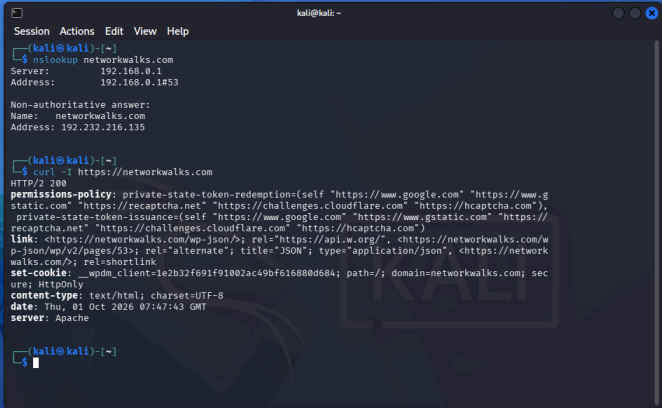
No active exploitation was performed during this exercise.

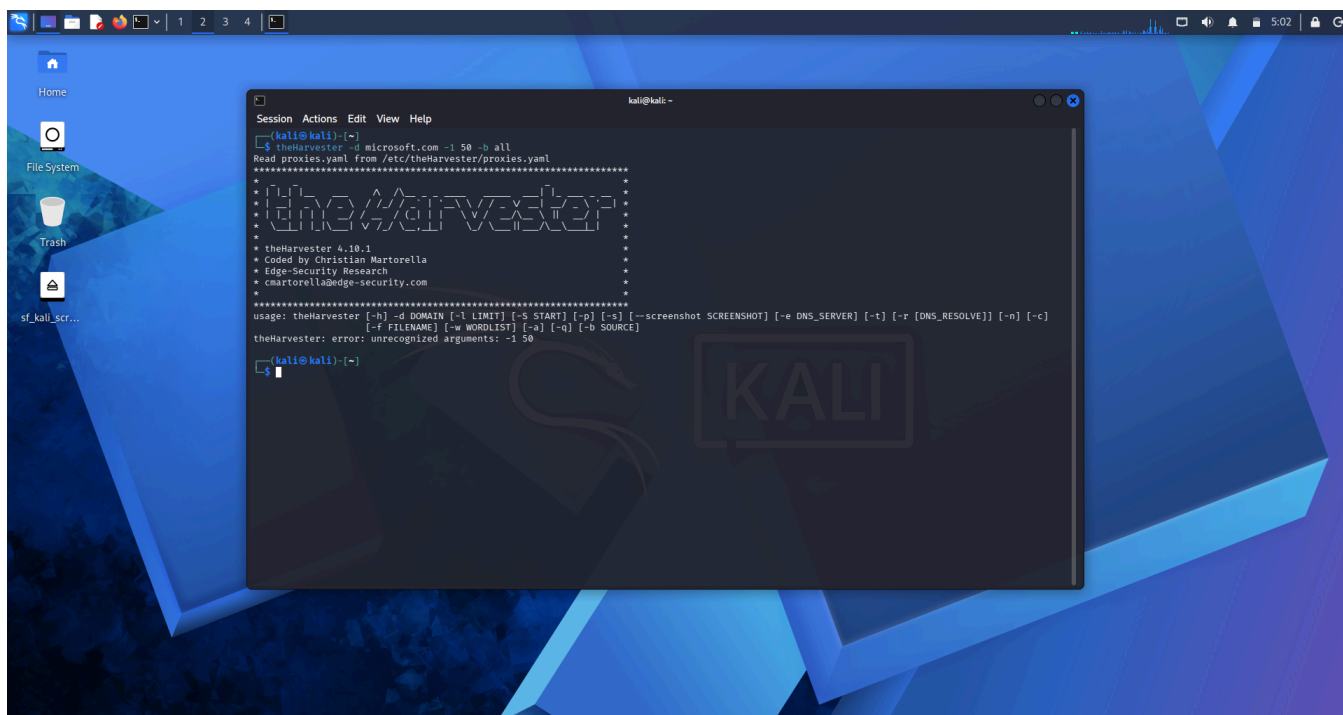
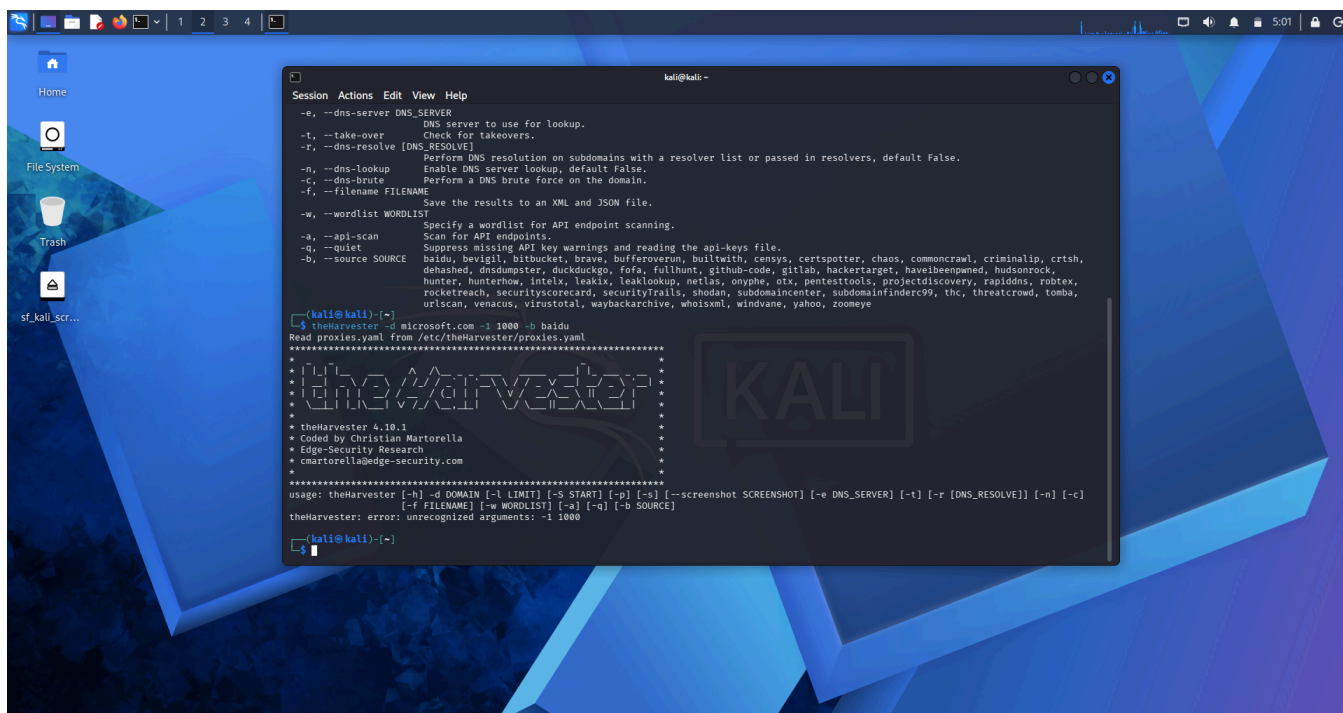
8. Evidences Collected



```
kali@kali: ~  
$ whois networkwalks.com  
Domain Name: NETWORKWALKS.COM  
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.godaddy.com  
Registrar URL: http://www.godaddy.com  
Updated Date: 2025-11-12T10:08:43Z  
Creation Date: 2019-11-06T22:51:46Z  
Registry Expiry Date: 2027-11-06T22:51:46Z  
Registrar: godaddy.com, LLC  
Registrar IANA ID: 146  
Registrar Abuse Contact Email: abuse@godaddy.com  
Registrar Abuse Contact Phone: 480-624-2105  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Name Server: NS6135.HOSTGATOR.COM  
Name Server: NS6136.HOSTGATOR.COM  
DNSSEC: unsigned  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/w
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-End-



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Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub